

Establishing a Standard Operating Procedure (SOP) for Palm Oil Plantation FFB Image Capture: Utilizing YOLOv8 for Counting and Ripeness Classification

1st Muhammad Rizky Hidayat
Computer Science Department
School of Computer Science
Bina Nusantara University
Jakarta, Indonesia 11480
muhammad.hidayat016@binus.ac.id

2nd Pavel Azarya Sulistyio
Computer Science Department
School of Computer Science
Bina Nusantara University
Jakarta, Indonesia 11480
pavel.sulistyio@binus.ac.id

3rd Hansen Oktario
Computer Science Department
School of Computer Science
Bina Nusantara University
Jakarta, Indonesia 11480
hansen.oktario@binus.ac.id

4th Alexander A S Gunawan
Computer Science Department
School of Computer Science
Bina Nusantara University
Jakarta, Indonesia 11480
aagung@binus.edu

5th Prasetyo Mimboro
Plant Division, Regional 2
PT. Perkebunan Nusantara IV
Medan, Indonesia 11480
prasetyo.mimboro@ptpn4.co.id

Abstract—Almost all oil palm harvest management is still done manually today, especially in the monitoring stage where oil palm farmers assess the ripeness of oil palm fresh fruit bunches (FFB) in the plantation. The monitoring stage, which is a crucial step, requires re-checking the ripeness of fresh fruit bunches (FFB) at the collection point (TPH) to ensure that the bunches brought to the factory are ripe. This research focuses on creating an SOP to be implemented in the plantation so that the AI in the automated system, also created in this research, can be used to optimally detect, calculate, and classify the ripeness of the FFB. By using the YOLO8 algorithm based on deep learning and CNN, the dataset used consists of 4257 images taken directly from PTPN IV plantations. The results of this study showed good model performance, with an accuracy of 96.56%. The application of this model is expected to increase the efficiency and accuracy of FFB quality monitoring, reduce losses, and support better data-based decision-making.

Keywords—palm oil, ripeness classification, deep learning, operational efficiency, harvest management

I. INTRODUCTION

In Indonesia, palm oil is the largest supplier of vegetable oil [1], [2]. The important role of palm oil in the Indonesian economy requires a tight operational flow of palm oil [2]. One of the state-owned companies involved in oil palm plantations is PTPN, which has a number of operations for its own harvesting activities. PTPN's harvesting operations are known as harvest management.

The main stages of harvest management in PTPN include planning, organizing, implementing and monitoring. In general, harvest management involves a series of activities ranging from cutting ripe bunches according to criteria, collecting and gathering loose fruit, and arranging bunches at the collection point (TPH). Currently, all harvest management activities are still done manually by the Harvest Foreman and the Afdeling Assistant. The Harvest Foreman is in charge of making all the preparations before the harvest day, such as recording and estimating the resources needed, while the Afdeling Assistant has full control over the Afdeling's harvesting activities according to the plan prepared.

The monitoring phase is a crucial step. This step consists of calculating and rechecking the eligibility of the FFBs at the TPH. This verification is mandatory as it has a direct impact on the quality of the crude palm oil if any raw bunches are transported to the factory [1]-[6]. This step can take a long time. If there are problems, such as bad weather or human error, there will be delays in all the harvest management steps. These delays can result in block-by-block harvesting not being completed, time conflicts when FFB is transported from TPH multiple times, potential resting time, trucks having to wait and turn around until they are full, damaged roads due to being passed more than once in a harvesting round, idle capacity at the mill, and increased harvesting costs.

After a series of activities at the TPH, the FFBs are loaded into the factory according to a predetermined schedule. Harvest management SOPs are of great importance, especially the criteria for the ripeness level of the FFB. This SOP has a direct impact on the processed crude oil because, in addition to preventing losses, the SOP on the ripeness level of the FFB affects the yield, quality and grade of crude palm oil (CPO) and palm kernel oil (PKO). Therefore, it is very important to classify the ripeness level of FFB and calculate the amount of FFB harvested at TPH before transportation to the palm oil mill to maintain the oil quality.

In this research, the researchers propose the YOLO (You Only Look Once) algorithm model used in computer vision [4]-[8] because it is considered to fit the needs of the field with the ease and lightness of the model in execution. To improve the performance of the model, this research collects a larger and more complex data set in accordance with the conditions in the field, using internal data belonging to PTPN IV. The researcher will study the proposed model with techniques to prevent overfitting and ensure accuracy in counting objects [4], [5], [8]. The model will be tested by photographing the bunches at the TPH to create an SOP for photographing. The ability of the model to detect objects, classify objects, and count objects according to their class will be the main concern. The way to take pictures that maximizes object detection results will be used as the SOP for arranging and photographing bunches with smartphones in this study.

II. LITERATURE REVIEW

A. Deep Learning

Deep learning is a subfield of machine learning that uses artificial neural networks to analyze data and make decisions based on patterns found in the data. Deep learning is particularly effective in image classification and object detection tasks due to its ability to learn complex and non-linear data representations [5], [15].

B. Convolutional Neural Networks (CNN)

CNN is a neural network architecture specifically designed for image processing tasks. CNNs consist of a convolution layer that extracts features from the input image, followed by a pooling layer that reduces the dimensionality of the data, and a fully connected layer that performs classification [9].

C. YOLO Algorithm

CNN is a neural network architecture specifically designed for image processing tasks. CNNs consist of a convolution layer that extracts features from the input image, followed by a pooling layer that reduces the dimensionality of the data, and a fully connected layer that performs classification [9].

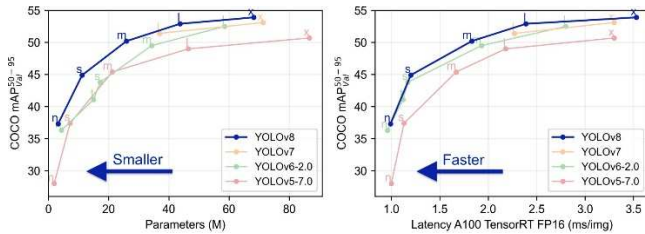


Fig. 1. Comparison graph of the latest generation 5 to 8 YOLOs [18]

YOLO (You Only Look Once) is a fast and efficient object detection algorithm capable of detecting objects in real time [13], [18]. The latest generation of YOLO, such as YOLOv8 (see Fig. 1), offers significant improvements in detection accuracy and speed over previous versions, making it an ideal choice for real-time applications [13], [18].

Models	Precision	Recall	mAP50	mAP50-95	F1
YOLOv8n	0.950	0.933	0.978	0.817	0.941
YOLOv8-SimSPPF	0.955	0.940	0.978	0.820	0.947
YOLOv8-GAM	0.962	0.921	0.978	0.819	0.941
YOLOv8-AKConv	0.958	0.959	0.983	0.846	0.958
YOLOv8-SimSPPF+AKConv	0.973	0.961	0.989	0.865	0.967
YOLOv8-AKConv+GAM	0.969	0.964	0.986	0.860	0.966
YOLOv8-SimSPPF+GAM	0.976	0.963	0.988	0.865	0.969
YOLOv8-AKConv+GAM+SimSPPF	0.966	0.971	0.991	0.878	0.968

Fig. 2. Performance comparison of YOLOv8 object detection models [13]

Based on (see Fig. 1), the graph illustrates a clear trend where increasing the various scales also increases the accuracy of the model [18]. Each model includes different scales denoted by n (*nano*), s (*small*), m (*medium*), l (*large*), and x (*extra-large*). YOLOv8n (Nano) (see Fig. 2) is a smaller, lighter version of the YOLOv8 model, optimized for speed and efficiency in resource-constrained environments such as edge devices or mobile applications. In other words, YOLOv8n is designed to run faster because of its smaller architecture. The researchers chose this model because it is lightweight for real-time bunch detection, especially to compensate for the lack of mobile device specifications in the field.

D. Transfer Learning

Transfer learning (TL) is a technique in machine learning where a model trained for one task can be used to solve another task. This technique can speed up training time and increase

efficiency so that the model's performance on different tasks can improve [24]. The method used is pre-training, where researchers use a model that has been trained on the same or similar task as the new task, but then use the model as a starting point to build a new model from scratch.

E. Object Detection in Agriculture

In the palm oil industry, object detection technology can be used to automate the process of monitoring and grading FFB at the TPH. Companies can improve the accuracy of FFB ripeness classification, reduce losses and improve operational efficiency. Previous research shows that the use of this technology can provide significant benefits in improving the quality and quantity of oil palm production [3]-[6].

F. Image Annotation and Dataset Preparation

The creation of a high quality dataset is an important step in the development of an object recognition model [3]. This process involves image annotation, where objects in the image are labeled and bounding boxes are used to train the model. A good dataset should cover a wide range of lighting conditions, viewing angles, and object variations to ensure that the model performs well in different real-world situations [3], [18].

G. Evaluation Metrics for Object Detection

The performance evaluation of object detection models is usually done using metrics such as mean average precision (mAP), intersection over union (IoU), and inference time. These metrics help to measure the accuracy, precision, and speed of the model in detecting and classifying objects [7], [18]. The use of appropriate evaluation metrics is essential to ensure that the developed model can meet the needs of real-world applications.

III. PROPOSED METHOD

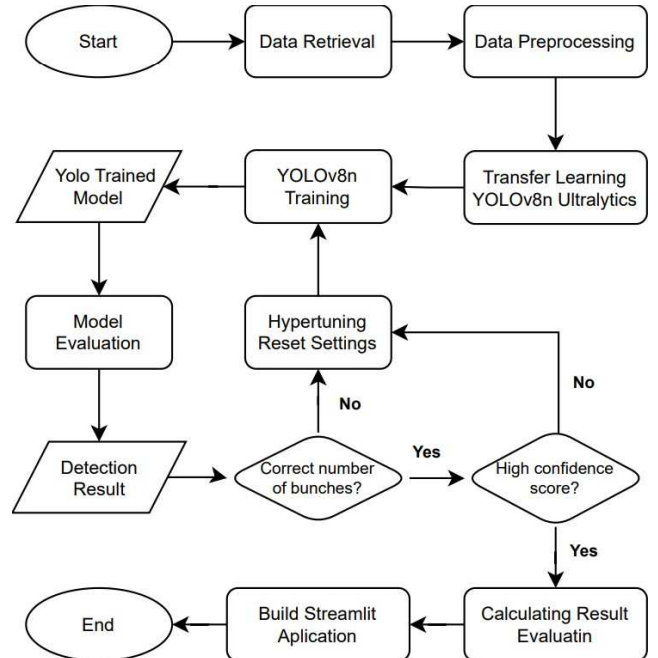


Fig. 3. The methodology of this research

A. Data Retrieval

The data was taken from the oil palm plantation of PTPN IV. The data taken is an image of a collection of oil palm bunches in the TPH, with a total of 4257 images collected.

B. Data Preprocessing

This stage consists of cleaning, resizing, annotation, and augmentation. By resizing all images to the input size expected by the YOLO model, then labeling each object in the image with bounding boxes according to its classification, and finally performing transformations such as rotation, flipping, and lighting changes to increase data variation.



Fig. 4. Class balance of all bounding box data after data augmentation

The bounding boxes of the raw and the deformed data may be underrepresented, but this comparison does not affect the training specifically. It is rare to find raw and deformed bunches due to the harvesting policy that requires only ripe bunches to be harvested.



Fig. 5. Setting the data split ratio

C. Model Development

The research continued by performing a transfer learning model from Ultralytics' pre-trained weights to speed up the training process and improve performance.

```
%pip install -q ultralytics
%cd ultralytics
%pip install -U ultralytics
```

Fig. 6. Loads pre-trained weights ultralytics

Using the pre-trained weights on Ultralytics, the model was retrained using new preprocessing data to match the specific task [24] of detecting and classifying FFB in images with a new dataset from PTPN IV by using Pre-trained Weights and Fine-tuning.

```
!yolo task = detect \
mode = train \
model = yolov8n.pt \
data = /content/drive/MyDrive/YOLO8/chip.yaml \
epochs = 200 \
workers = 16 \
batch = -1 \
imgsz = 800 \
save = True \
cache = ram \
verbose = True \
profile = True \
val = True \
plots = True
```

Fig. 7. Hypertuning parameter settings

Task: *detect*, sets the task for object detection. Mode: *train*, indicates that the model is being trained. Model: *yolov8n.pt*, specifies the nano version of the YOLOv8 model used. Data: Path to the data set configuration file (chip.yaml).

Epochs: 200, sets the number of epochs for training. Workers: 16, the number of workers used to load data. Batch: -1, use an automatic batch size. Imgsz: 800, sets the input image size to 800x800 pixels. Save: *True*, saves the training results. Cache: *ram*, caches the data set in RAM for faster access. Verbose, Profile, Val, Plots: *True*, enables detailed log, profile, validation, and plotting of training results.

These parameters are designed to optimize the training process of the YOLOv8 model, focusing on efficiency and quality of the recognition results. As with the Batch Processing setting, data is processed in batches for computational efficiency. Setting Batch Processing to -1 allows the model to process multiple tasks at once, depending on the resource limits of the computer, speeding up training and taking advantage of parallelism on hardware such as a GPU.

D. Build Streamlit Application

The implementation of the model in this research uses webapps with the Streamlit framework, which is used to build an interactive user interface for data science applications by simplifying web application development using Python.

```
Data: An uploaded image
Result: Annotated image with counted objects
Load the YOLO model from the specified path;
Define class names and corresponding colors;
Display the title 'Palm Oil Detection and Counting';
Provide an option to upload an image;
Provide an option to choose detection mode
(Bunches <= 30 or Bunches > 30);
Set confidence threshold based on the selected mode;
if an image is uploaded then
    Read the uploaded image;
    Display the uploaded image with a caption;
    Convert the image from PIL format to OpenCV
    format;
    Perform object detection using the YOLO model
    with the specified confidence threshold;
    Initialize counters for Total, Ripe, Raw, and
    Deformed to 0;
    Copy the image to create an annotated version;
    foreach result in detection results do
        Extract bounding boxes and class indices;
        foreach class index in class indices do
            Increment the Total counter;
            if class index is 0 then
                Increment the Ripe counter;
            end
            else if class index is 1 then
                Increment the Raw counter;
            end
            else if class index is 2 then
                Increment the Deformed counter;
            end
            Extract coordinates of the bounding box;
            Get the corresponding label and color;
            Create the label with the order number;
            Draw the bounding box on the image;
            Draw the background for the text;
            Put the label with the order number on the
            image;
        end
    end
    Convert the annotated image from OpenCV format
    to PIL format;
    Display the annotated image with a caption;
    Display the total count and counts of each category
    (Ripe, Raw, Deformed);
end
```

Fig. 8. Pseudocode for calculating objects from the bounding box detected by YOLOv8 with Streamlit

E. Model Result Evaluation

The results are presented in an image containing the bounding box of each detected bunch and the number of labels of each class on the label. The detected images are evaluated to ensure the accuracy and precision of the detection and classification model under actual field conditions.

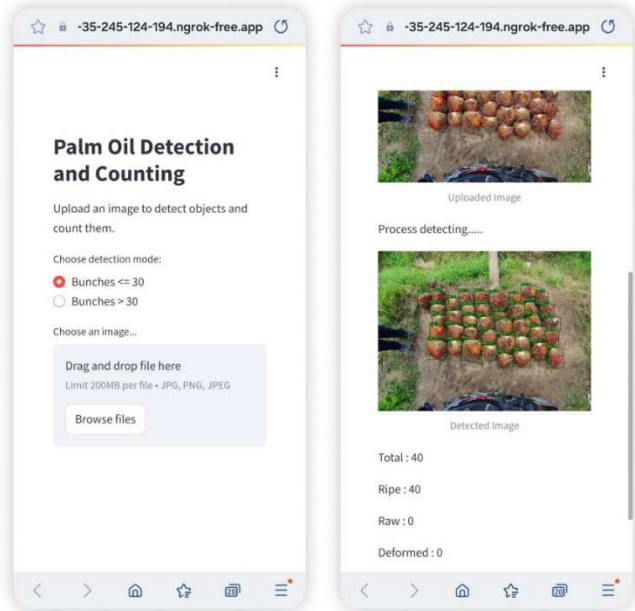


Fig. 9. Web application interface using Streamlit

Of course, there will be challenges in implementing the model in plantations, such as environmental factors, sensor limitations, and integration with existing plantation management systems. The results of observations in the field by researchers show that the environment is friendly in the sense that it is very supportive for harvesting and workers who carry complete tools with smart and latest mobile devices in the sense that they are sophisticated enough for communication and daily life. Therefore, these challenges can be considered as answered with the existing conditions.

IV. RESULTS AND DISCUSSION

A. Performance Metrics Deep Dive

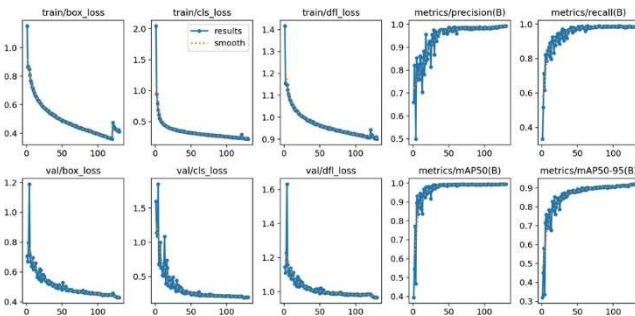


Fig. 10. YOLOv8n's best set of training result graphs

The process of training the YOLOv8 model using the collected and augmented data resulted in several important performance metrics. The YOLOv8 model has optimal performance in detecting and classifying oil palm FFBs. The training box loss, classification loss, and distance field loss (DFL) all decrease consistently, indicating the model's

improved ability to accurately predict the bounding box and correctly classify objects. The validation loss also follows the same trend, indicating that the model generalizes well and can classify unseen data well. The precision and recall metrics both show a steady increase and reach high values. This means that the model's predictions are becoming more accurate and it is effectively detecting most of the relevant objects. The mAP50 metric shows a high and increasing trend, indicating good performance at the IoU threshold of 50%.

The precision, recall, mAP50, and mAP50-95 metrics all increase steadily, indicating the improved accuracy and performance of the model in detecting relevant objects at different IoU thresholds. Overall, the consistent decrease in all loss metrics during training and validation, along with the high precision, recall, and mAP values, indicate that the model performs well without overfitting. It effectively detects and classifies objects and maintains good performance on both training and unseen validation data.

B. Confusion Matrix

		TRUE			
		Ripe	Raw	Deformed	Background
PREDICT	Ripe	132	9	0	10
	Raw	2	1189	0	28
	Deformed	0	0	474	10
	Background	0	5	0	0

Fig. 11. Confusion matrix training results

Accuracy is calculated by dividing the number of correct predictions by the total number of predictions. In this case, the model successfully predicted 96.56% of the samples correctly. Thus, the learning model's ability to guess the bunch ripeness criteria in the image has achieved high accuracy and minimal errors.

The background column in the table is the Binary Cross Entropy (BCE) or object loss in YOLO, which is calculated using the Binary Cross Entropy. It is intended to help the model determine whether a portion of the image contains an object or not. This is especially important in object detection tasks, where the model must not only detect the object, but also determine its location in the image. If the YOLO model is not trained to recognize the background, then any part of the image will be considered a potential object. This will lead to inaccurate predictions because the model cannot clearly distinguish relevant objects from noise or unimportant elements in the image, and the model will produce many false positives because it considers the background to be an object.

C. Proposed Standard Operating Procedure (SOP)



Fig. 12. Sample dataset received by the researcher from PTPN IV

Image (a) and image (b) are the same stack of bunches, but they differ in the shooting distance. This difference in distance affects the accuracy of the model in detecting objects and predicting ripeness. Image (a) taken from a closer distance results in more accurate detection than image (b). There is no limit to the number of bunches that can be detected by the model, depending on the angle of the image and the lighting. Figure (f) shows the correct image detection result and is an example of excellent image detection.

It can be concluded that distance, lighting and shooting angle affect the object detection results, especially in images with a large number of objects. For example, images (d) and (e) have many objects, but only image (e) has convincing prediction and detection results. The best Standard Operating Procedure (SOP) for taking images is to have sufficient lighting, a shooting angle between 45° - 90° , bunches below 30, and a distance that is not too small or cropped.

Based on (see Fig. 11), two features are created that can be modified to improve object prediction. The creation of these two features is done by differentiating the confidence level. If there are many objects, then the confidence used is 40% and vice versa for 60%. This is done so that objects that look small can be detected.

V. CONCLUSION AND FUTURE WORKS

Overall, the results show that the trained YOLOv8 model successfully improved its performance on both the training and validation data. The (see Fig. 10) and (see Fig. 11) confirm that the model is able to learn effectively and generalize well to previously unseen data. With these results, the trained YOLOv8 model can be implemented in a computer vision-based object detection and counting system at TPH PTPN IV. This implementation is expected to improve the efficiency and accuracy of FFB quality monitoring, as well as support better decision making based on accurate and real-time data.

Future research could focus on improving the performance of the YOLOv8 model by incorporating larger and more diverse data sets, further physiological and contextual features, and optimized hyperparameters, or by using the latest version of the YOLO model. By discussing these research directions, this paper can advance the crop management process and develop more effective prevention, intervention, and support strategies to prevent yield loss. Ultimately, these efforts will contribute to a more optimized harvest that is safer and more inclusive to maximize company production.

AUTHOR CONTRIBUTION

This research was planned by Alexander Agung Santoso Gunawan. Experiments and paper writing were conducted by Muhammad Rizky Hidayat with the guidance of Alexander Agung Santoso Gunawan. References and literature research were collected and reviewed by Pavel Azarya Sulisty and Hansen Oktario. Research data support was provided by Prasetyo Mimboro. All authors have read and approved the manuscript.

AUTHOR CONTRIBUTION

This research shares the code, deep learning model, and application publicly on GitHub. You can access them at <https://github.com/kaykyysee/PalmOilClassification.git>.

REFERENCES

- [1] R. A. Aljawadi, D. Ahmad, N. M. Nawi, and M. Saufi, "Mechanized harvesting of oil palm fresh fruit bunches: A review," in Proc. 7th Kuala Lumpur Int. Agric. For. Plant. Conf. (KLIAFP7), Bangi, Malaysia, Dec. 2018.
- [2] H. A. Hasibuan, "Determination of Yield, Quality and Chemical Composition of Palm Oil and Palm Kernel Oil of Fresh Fruit Bunches with Variation ripeness as a Basic for Determining Harvest ripeness Standard," J. Penelit. Kelapa Sawit, vol. 27, no. 3, pp. 123-132, 2020.
- [3] Suharjito, F. A. Junior, Y. P. Koeswandy, Debi, P. W. Nurhayati, M. Asrol, and Marimin, "Annotated Datasets of Oil Palm Fruit Bunch Piles for Ripeness Grading Using Deep Learning," Scientific Data, vol.10, no.72, 2023.
- [4] Suharjito, M. Asrol, D. N. Utama, F. A. Junior, and Marimin, "Real-Time Oil Palm Fruit Grading System Using Smartphone and Modified YOLOv4," in *IEEE Access*, vol. 11, no. Y, pp. 59758 - 59773, 2023.
- [5] Junos, M. H., Khairuddin, A. S. M., Thannirmalai, S., & Dahari, M., "An optimized YOLO-based object detection model for crop harvesting system," IET Image Processing, vol.15, no.9, pp.2112-2125, 2021.
- [6] A. Y. Saleh and E. Liansitim, "Palm oil classification using deep learning," Science in Information Technology Letters, vol. 1, no. 1, pp. 1-8, May 2020.
- [7] J. Qiu, J. Liu, and Y. Shen, "Computer Vision Technology Based on Deep Learning," in Proc. 2021 IEEE 2nd Int. Conf. Inf. Technol., Big Data, Artif. Intell. (ICIBA), Chongqing, China, Dec. 2021.
- [8] F. Fuentes-Penailillo et al., "Automating Seedling Counts in Horticulture Using Computer Vision and AI," Hort., vol. 9, no. 10, p. 1134, Oct. 2023.
- [9] S. Albawi, T. A. Mohammed, and S. Al-Zawi, "Understanding of a Convolutional Neural Network," in 2017 International Conference on Engineering and Technology (ICET), Antalya, Turkey, 2017, pp. 1-6.
- [10] A. Bhargava, A. Shukla, O. P. Goswami, M. H. Alsharif, P. Uthansakul, and M. Uthansakul, "Plant Leaf Disease Detection, Classification, and Diagnosis Using Computer Vision and Artificial Intelligence: A Review," IEEE Access, vol. 12, pp. 37443-37469, 2024.
- [11] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. Cambridge, MA: MIT Press, 2016.
- [12] X. Ji, J. Henriques and A. Vedaldi, "Invariant Information Clustering for Unsupervised Image Classification and Segmentation," in *arXiv preprint arXiv:1807.06653*, 2018.
- [13] X. Qin, C. Yu, B. Liu and Z. Zhang, "YOLO8-FASG: A High-Accuracy Fish Identification Method for Underwater Robotic System," in *IEEE Access*, vol.11, 2023.
- [14] S. Belciug, "Autonomous Fetal Morphology Scan: Deep Learning Clustering and Merge," in *BMC Med. Inform. Decis. Mak.*, vol. 24, no. 102, 2024.
- [15] X. Li and Y. Shi, "Computer Vision Imaging Based on Artificial Intelligence," in Proc. 2018 Int. Conf. Virtual Reality Intelligent Syst. (ICVRIS), 2018, pp.22-25.
- [16] A. Mumuni, F. Mumuni, and N. K. Gerrar, "A Survey of Synthetic Data Augmentation Methods in Machine Vision," Machine Intelligence Research, 2024.
- [17] X. Song, S. Cao, J. Zhang, and Z. Hou, "Steel Surface Defect Detection Algorithm Based on YOLOv8," Electronics, vol.13, no.988, Mar. 2024.
- [18] J. Terven, D.-M. Cordova-Esparza, and J.-A. Romero-Gonzalez, "A Comprehensive Review of YOLO Architectures in Computer Vision: From YOLOv1 to YOLOv8 and YOLO-NAS," Mach. Learn. Knowl. Extr., vol. 5, pp. 1680-1716, Nov. 2023.
- [19] M. Majumder and C. Wilmot, "Automated Vehicle Counting from Pre-Recorded Video Using You Only Look Once (YOLO) Object Detection Model," J. Imaging, vol.9, no.131, pp.1-19, 2023.
- [20] T. A. R. Shyaa and A. A. Hashim, "Enhancing Real Human Detection and People Counting Using YOLOv8," in Proceedings of the BIO Web of Conferences, vol. 97, no. 00061, pp. 1-8, 2024.

- [21] Z. Zhong, "Pedestrian Detection and Gender Recognition Utilizing YOLO and CNN Algorithms," in Proc. 2023 Int. Conf. Mach. Learn. Autom., 2023, pp. 133-136.
- [22] C.-C. Hsieh, T.-Y. Hsu, and W.-H. Huang, "An Online Rail Track Fastener Classification System Based on YOLO Models," Sensors, vol. 22, no. 24, 2022.
- [23] M. Kristo, M. Ivasic-Kos, and M. Pobar, "Thermal Object Detection in Difficult Weather Conditions Using YOLO," in *IEEE Access*, vol. 8, pp. 125459-125476, 2020.
- [24] Q. Shi, Y. Zhang, X. Liu, and K. Zhao, "Regularised transfer learning for hyperspectral image classification," IET Comput. Vis., vol. 13, no. 2, pp. 188-193, Feb. 2019.